Course Type	Course Code	Course Name	Credits
PEC	CEPEC7041	APPLIED DATA SCIENCE	03

Examination Scheme					
Distribution of Marks			Exam Duration (Hrs.)		Total Marks
In-semester Assessment		End Semester Examination (ESE)			
Continuous Assessment	Mid-Semester Exam (MSE)		MSE	ESE	
20	30	50	1.5	2	100

Pre-requisite:

1. CEPEC5011-Data Mining & Business Intelligence
2. CEPEC6021-Machine Learning

Program Outcomes addressed:

1. PO1: Engineering knowledge.
2. PO2: Problem analysis.
3. PO3: Design/development of solutions.
4. PO5: Modern tool usage
5. PO11: Life-long learning.

Course Objectives:

1	To introduce students to the fundamental concepts of Data Science, its process, and its applications across domains.
2	To enable learners to perform data exploration using descriptive statistical techniques and visual analytics.
3	To develop students' ability to apply inferential statistical methods for drawing meaningful conclusions from data.
4	To train learners in regression modelling, evaluation of model performance, and interpretation of predictive relationships.
5	To facilitate students' understanding of anomaly detection techniques and their application in identifying unusual patterns in data.
6	To equip learners with the knowledge and skills required for analysing and forecasting time-dependent data using time series models.

Module	Details	Hrs
	Course Introduction Applied Data Science provides a comprehensive understanding of the concepts, tools, and methodologies used to interpret data, identify patterns, and support evidence-based decision-making. The course introduces learners to the complete data science workflow, encompassing data acquisition, preparation, exploration, modelling, and evaluation. It emphasizes essential statistical techniques, regression methods, anomaly detection approaches, and time series forecasting to illustrate how analytical insights are generated from real-world datasets. By integrating conceptual understanding with practical relevance, the course equips students with the analytical skills necessary for advanced study and application in fields such as machine learning, artificial intelligence, and data-driven research.	**01**

01.	**Introduction to Data Science** ***Learning Objectives:*** *To introduce the fundamental concepts of Data Science, its process, and the motivation for using data-driven techniques.*	**04**
	Contents: 1.1 Introduction to Data Science, Data Science Process 1.2 Motivation to use Data Science Techniques: Volume, Dimensions and Complexity, Data Science Tasks and Examples 1.3 Overview of Data Preparation, 1.4 Model building, Cross Validation, K-fold cross validation, leave-1 out, Bootstrapping 1.5 Difference between data science and data analytics. 1.6 Applications of Data Science	
	Self-Learning Topics: *Roles in Data Science: Data Engineer, ML Engineer, Analyst. Real-world Case Studies (Healthcare, Finance, Transportation, Recommender Systems).*	
	Learning Outcomes: *A learner will be able to* *LO 1.1 - Apply the knowledge of discrete structures, linear algebra, statistics, and numerical techniques to interpret fundamental data characteristics such as volume, dimensionality, and complexity. (PI 1.1.1)* *LO 1.2-Apply engineering fundamentals to explain the core concepts of Data Science, its overall process, and the motivation for adopting data-driven approaches. (PI 1.3.1)* *LO 1.3- Evaluate a given data-related problem statements and determine the required analysis, while continually learning and adapting to new data science tools and techniques. (PI 2.1.1, PI 11.1.1)* *LO 1.4- Apply computer engineering principles to outline appropriate data science procedures and basic model evaluation methods such as cross-validation, K-fold, and bootstrapping, while remaining aware of ongoing technological advancements in the field. (PI 2.3.1, PI 11.2.2)*	
02.	**Data Exploration using Descriptive Statistics** ***Learning Objective/s:*** *To provide learners with the knowledge to explore and summarize data using descriptive statistics, and to cultivate the ability to interpret patterns through appropriate graphical methods.*	**12**
	Contents: 2.1 Types of data, Properties of data 2.2 Statistical data–Classification, Tabulation, Diagrammatic & Graphic representation of data 2.3 Univariate Exploration: Measure of Central Tendency, Measure of Spread, Symmetry, Skewness: Karl Pearson Coefficient of skewness, Bowley 's Coefficient, Kurtosis 2.4 Multivariate Exploration: Central Data Point, Correlation, Different forms of correlation, Karl Pearson Correlation Coefficient for bivariate distribution. 2.5 Data Visualization Univariate Visualization: Histogram, Quartile, Distribution Chart Multivariate Visualization: Scatter Plot, Scatter Matrix, Bubble chart.	

	Self-Learning Topics: *Shape of distributions: heavy-tailed, light-tailed, multimodal distributions. Advanced visualizations: Violin Plot, Boxen Plot, Pair plot, Heatmaps.*	
	Learning Outcomes: *A learner will be able to* *LO 2.1-Apply the knowledge of statistics and numerical techniques to classify, tabulate, and summarize different types and properties of data. (PI 1.1.1)* *LO 2.2-Apply the concepts of probability and statistics to compute and interpret measures of central tendency, dispersion, skewness, kurtosis, and correlation.* *(PI 1.1.2)* *LO 2.3-Evaluate a given dataset to identify key objectives by examining univariate and multivariate characteristics using appropriate statistical measures. (PI 2.1.1)* *LO 2.4-Analyze and interpret data patterns using contemporary visualization tools through histograms, scatter plots, scatter matrices, bubble charts, and other graphical representations. (PI 2.4.2)* *LO 2.5-Identify suitable modern tools and techniques for performing univariate and multivariate data visualization. (PI 5.1.1)* *LO 2.6-Demonstrate proficiency in using appropriate data visualization tools to create histograms, scatter plots, bubble charts, and other graphical representations. (PI 5.2.2)*	
03.	**Data Exploration using Inferential Statistics** ***Learning Objective/s:*** *To provide learners with the knowledge to apply inferential statistical methods for drawing conclusions from data, and to cultivate the ability to select appropriate hypothesis tests for valid decision-making.*	**06**
	Contents Overview of Various forms of distributions: Normal, Poisson, Test Hypothesis, Central limit theorem, Confidence Interval, Z-test, t-test, Type-I, Type-II Errors, ANOVA	
	Self-Learning Topics: *Neyman-Pearson lemma-MP and UMP tests. Practical examples using Python/Excel/R.*	
	Learning Outcomes: *Learner should be able to* *LO 3.1-Apply the knowledge of statistics and numerical techniques to describe and analyse probability distributions such as normal and Poisson distributions. (PI 1.1.1)* *LO 3.2-Apply the concepts of probability and statistics to compute confidence intervals and perform hypothesis tests such as Z-test, t-test, and ANOVA. (PI 1.1.2)* *LO 3.3-Identify appropriate statistical methods and hypothesis testing procedures by determining the mathematical requirements of the given problem. (PI 2.1.3)* *LO 3.4-Arrive at valid statistical conclusions by interpreting test statistics, p-values, confidence levels, and the implications of Type-I and Type-II errors. (PI 2.4.4)*	
04.	**Regression Analysis** ***Learning Objective/s:*** *To provide learners with the knowledge to build and interpret regression models, and to cultivate the ability to evaluate model performance using appropriate statistical measures.*	**10**

	Contents: 4.1 Mathematical and Statistical Equation – Meaning of Intercept and Slope 4.2 Error term – Measure for Model Fit –R2 – MAE – MAPE. 4.3 Multiple Linear Regression Model, 4.4 Partial Regression Coefficients, 4.5Testing Significance overall significance of Overall fit of the model, 4.6 Testing for Individual Regression Coefficients	
	Self-Learning Topics: *Residual analysis and its importance in regression diagnostics. Real regression case studies (house prices, sales forecasting).*	
	Learning Outcomes: *A learner will be able to* *LO 4.1-Apply the knowledge of statistics and numerical techniques to construct regression equations and interpret parameters such as intercept, slope, and error term. (PI 1.1.1)* *LO 4.2-Apply the concepts of probability and statistics to compute model fit metrics such as R^2, MAE, and MAPE, and interpret their significance. (PI 1.1.2)* *LO 4.3-Identify appropriate mathematical and statistical methods required for selecting suitable regression models such as simple, multiple, and partial regression. (PI 2.1.3)* *LO 4.4-Analyze and interpret regression model outcomes using contemporary tools to assess significance of coefficients and overall model performance. (PI 2.4.2)* *LO 4.5-Explore alternative regression model designs by examining different model structures and variable selections for meeting analytical requirements. (PI 3.2.1)* *LO 4.6-Perform systematic evaluation of regression models by comparing fit statistics, significance tests, and error measures to select the optimal design. (PI 3.3.1)*	
05.	**Anomaly Detection** ***Learning Objective/s:*** *To provide learners with the knowledge to identify and detect anomalies in data, and to cultivate the ability to apply appropriate statistical and distance-based techniques for recognizing unusual patterns.*	**06**
	Contents: 5.1 Outliers, Causes of Outliers, 5.2 Anomaly detection techniques, 5.3 Outlier Detection using Statistics 5.4 Outlier Detection using Distance based method, 5.5 Outlier detection using density-based methods, 5.6 SMOTE	
	Self-Learning Topics: *Introduction to autoencoders for anomaly detection, Performance metrics for anomaly detection (AUC-PR, F1, confusion matrix).*	

	Learning Outcomes: *A learner will be able to* *LO 5.1-Apply statistical knowledge to explain the concept of outliers and identify their possible causes in real-world datasets. (PI 1.1.1)* *LO 5.2 -Apply probability and statistical techniques to detect anomalies using distribution-based and statistical thresholding methods. (PI 1.1.2)* *LO 5.3 -Identify appropriate mathematical and algorithmic approaches for detecting outliers using distance-based and density-based methods. (PI 2.1.3)* *LO 5.4-Analyze and interpret results of anomaly detection techniques using contemporary analytical tools and visualizations. (PI 2.4.2)* *LO 5.5-Explore alternative anomaly detection designs, including statistical, distance-based, and density-based techniques, to address varied data scenarios. (PI 3.2.1)* *LO 5.6 -Perform systematic evaluation of anomaly detection methods, including SMOTE for class imbalance, to determine the most effective solution. (PI 3.3.1)* *LO 5.7 -Identify suitable modern engineering and data science tools required for implementing anomaly detection algorithms effectively. (PI 5.1.1)* *LO 5.8 -Demonstrate proficiency in using discipline-specific tools for implementing anomaly detection techniques and interpreting their outputs. (PI 5.2.2)*	
06.	**Time Series Forecasting** ***Learning Objective/s:*** *To provide learners with the knowledge to analyze and model time-dependent data, and to cultivate the ability to apply suitable forecasting techniques for predicting future trends.*	**05**
	Contents: 6.1 Taxonomy of Time Series Forecasting methods, Time Series Decomposition 6.2 Smoothening Methods: Average method, Moving Average smoothing, Time series analysis using linear regression, ARIMA Model, 6.3 Performance Evaluation: Mean Absolute Error, Root Mean Square Error, Mean Absolute Percentage Error, Mean Absolute Scaled Error	
	Self-Learning Topics: *Seasonal decomposition, Stationarity concept and tests.*	
	Learning Outcomes: *A learner will be able to* *LO 6.1 -Apply statistical and numerical techniques to interpret the taxonomy of time series forecasting methods and the components of time series decomposition such as trend, seasonality, and irregular variations. (PI 1.1.1)* *LO 6.2 -Apply concepts of probability and statistics to implement smoothing methods such as the average method, moving average smoothing, and exponential approaches for analysing time-dependent data. (PI 1.1.2)* *LO 6.3 -Identify appropriate mathematical and statistical methods, including linear regression and ARIMA, required for selecting suitable time series forecasting models. (PI 2.1.3)* *LO 6.4-Analyze and interpret time series model outputs using contemporary analytical tools to understand forecast patterns and model behaviour. (PI 2.4.2)* *LO 6.5-Identify modern analytical and visualization tools required for building, analysing, and validating time series forecasting models. (PI 5.1.1)* *LO 6.6-Demonstrate proficiency in using discipline-specific tools to implement time series models, generate forecasts, and visualize predictions effectively. (PI 5.2.2)*	
	Course Conclusion The Applied Data Science course concludes by equipping learners with a solid understanding of how data-driven insights are derived through systematic statistical and analytical methods. Students gain the ability to explore and interpret datasets, perform valid inferential analysis, build and assess predictive and forecasting models, and detect	01

	anomalies in real-world data. By integrating foundational theory with practical analytical skills, the course prepares learners to apply evidence-based reasoning to complex problems and establishes a strong platform for advanced study in machine learning, artificial intelligence, and data-centric research.	
Total		**45**

Performance Indicators:

P.I. No.	P.I. Statement
1.1.1	Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems
1.1.2	Apply the concepts of probability, statistics and queuing theory in modelling of computer-based system, data and network protocols.
1.3.1	Apply engineering fundamentals
2.1.1	Evaluate problem statements and identifies objectives
2.1.3	Identify mathematical algorithmic knowledge that applies to a given problem
2.3.1	Able to apply computer engineering principles to formulate modules of a system with required applicability and performance.
2.4.2	Analyse and interpret the results using contemporary tools.
2.4.4	Arrive at conclusions with respect to the objectives.
3.2.1	Able to explore design alternatives.
3.3.1	Able to perform systematic evaluation of the degree to which several design concepts meet the criteria.
5.1.1	Identify modern engineering tools, techniques and resources for engineering activities
5.2.2	Demonstrate proficiency in using discipline-specific tools
11.1.1	Describe the rationale for the requirement for continuing professional development
11.2.2	Recognize the need and be able to clearly explain why it is vitally important to keep current regarding new developments in your field

Course Outcomes: Learner will be able to

1 Interpret data characteristics, understand core data science foundations, analyse data problems, and outline basic analytical procedures using emerging tools and techniques. (*LOs: 1.1,1.2,1.3,1.4*)

2 Analyse univariate and multivariate data patterns using statistical and visualization techniques, and apply modern tools to summarize, interpret, and effectively represent data. (*LOs:2.1,2.2,2.3,2.4,2.5,2.6)*

3 Evaluate statistical evidence using probability distributions and hypothesis testing, and apply appropriate statistical methods to compute confidence intervals, interpret test outcomes, and draw valid conclusions. (*LOs:3.1,3.2,3.3,3.4)*

4 Assess regression models using fit metrics, significance tests, and error measures, and apply statistical and computational techniques to construct, interpret, and select appropriate regression designs for data-driven problems. *(LOs:4.1,4.2,4.3,4.4,4.5,4.6)*

5 Analyse anomaly detection methods using statistical, distance-based, density-based, and sampling-based techniques, and apply analytical and computational tools to identify, interpret, and implement effective outlier detection solutions. (*LOs:5.1,5.2,5.3,5.4,5.5,5.6,5.7,5.8)*

6 Analyse time series forecasting methods and model behaviours using smoothing techniques, regression-based and ARIMA approaches, and apply modern analytical tools to build, interpret, and validate forecast models effectively. (*LOs:6.1,6.2,6.3,6.4,6.5,6.6)*

CO-PO Mapping Table with Correlation Level

CO ID	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CEPEC7041.1	3	3	--	--	--	--	--	--	--	--	3
CEPEC7041.2	3	3	--	--	3	--	--	--	--	--	--
CEPEC7041.3	3	3	--	--	--	--	--	--	--	--	--
CEPEC7041.4	3	3	3	--	--	--	--	--	--	--	--
CEPEC7041.5	3	3	3	--	3	--	--	--	--	--	--
CEPEC7041.6	3	3	--	--	3	--	--	--	--	--	--
Average	3	3	3	--	3	--	--	--	--	--	3

Text Books:

1. Vijay Kotu, Bala Deshpande. Data Science Concepts and Practice‖, Elsevier, M.K. Publishers.
2. Agarwal, B.L. (2006): Basic Statistics. Wiley Eastern Ltd., New Delhi
3. Gupta, S. P. (2011): Statistical Methods. Sultanchand &Sons, New Delhi

Reference Books:

1. Sivathanupillai, M &Rajagopal, K. R. (1979):-Statistics for Economics Students
2. Dr. P. N. Arora, Sumeet Arora, S. Arora, Ameet Arora, ―Comprehensive Statistical Methods‖, S.Chand Publications, New Delhi.
3. S.C. Gupta, V. K. Kapoor ―Fundamentals of Mathematical Statistics‖, S. Chand and Sons, New Delhi.

Other Resources:

1. Coursera – "Data Science Specialization" by Johns Hopkins; covers statistics, R, visualization. https://www.coursera.org/specializations/jhu-data-science
2. Google Data Analytics Certificate by Industry-level understanding of data analysis workflow. https://www.coursera.org/professional-certificates/google-advanced-data-analytics
3. MIT OpenCourseWare https://ocw.mit.edu/courses/18-05-introduction-to-probability-and-statistics-spring-2022/

IN-SEMESTER ASSESSMENT (50 MARKS)

1. Continuous Assessment - Theory-(20 Marks)

Two Class test: 05 marks each
Course Project: 05 Marks
Regularity and active participation: 05 Marks.

2. Mid Semester Exam (30 Marks)

Mid semester examination will be based on 40% to 50 % syllabus.

END SEMESTER EXAMINATION (50 MARKS)

End Semester Examination will be based on syllabus coverage up to the Mid Semester Examination (MSE) carrying 20%-30% weightage, and the syllabus covered from MSE to ESE carrying 70%-80% weightage.